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**Lab 5: Implementation of CI/CD Pipeline Using GitHub Actions
and Jenkins**

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Lab Report

Objective

To implement a Continuous Integration and Continuous Deployment (CI/CD) pipeline using GitHub Actions and Jenkins for automating the build, testing, and deployment process.

Tools Required

- Git
- GitHub
- GitHub Actions
- Jenkins
- Java/Node.js (Sample Project)
- Visual Studio Code

Theory

Continuous Integration (CI) is the practice of automatically building and testing code whenever changes are pushed to a repository.

Continuous Deployment (CD) automates the release of tested code to a deployment environment. GitHub Actions and Jenkins are popular CI/CD tools that help automate software development workflows, improving code quality and reducing manual effort.

Procedure

1. Create a project and upload it to a GitHub repository.
2. Configure a GitHub Actions workflow using a YAML file.
3. Commit and push the workflow file to trigger automated execution.
4. Install and configure Jenkins.
5. Connect Jenkins to the GitHub repository.
6. Create a Jenkins pipeline and configure build steps.
7. Execute the pipeline and verify successful build, testing, and deployment.

GitHub Actions Workflow

```
name: CI Pipeline

on:
  push:
    branches: [main]

jobs:
  build:
    runs-on: ubuntu-latest

    steps:
      - uses: actions/checkout@v4

      - name: Set up Java
        uses: actions/setup-java@v4
        with:
          distribution: temurin
          java-version: '17'

      - name: Build Project
        run: echo "Build Successful"
```

Jenkins Pipeline Example

```
pipeline {
  agent any

  stages {
    stage('Build') {
      steps {
        echo 'Building Project...'
      }
    }

    stage('Test') {
      steps {
```

```
        echo 'Running Tests...'
    }
}

stage('Deploy') {
    steps {
        echo 'Deploying
Application...'
    }
}
}
```

Result

Successfully implemented a CI/CD pipeline using GitHub Actions and Jenkins to automate the build, testing, and deployment process.

Conclusion

The experiment demonstrated how GitHub Actions and Jenkins automate software development workflows through Continuous Integration and Continuous Deployment. Implementing CI/CD pipelines improves development efficiency, reduces manual intervention, and ensures faster, more reliable software delivery.